emp_name dept_name
Vijay IT
Vinay IT
- Note: Only employees with matching dept_id in the departments table are included.

2. LEFT OUTER JOIN (LEFT JOIN)
A LEFT JOIN returns all rows from the left table (employees), and matching rows from the right table (departments). If no match is found, NULL is returned for the right table.
Example:
```
SELECT e.emp_name, d.dept_name
FROM employees e
LEFT JOIN departments d ON e.dept_id = d.dept_id;
```
Explanation:
This query retrieves all employees and their corresponding department names. If an employee doesn't belong to any department, NULL is returned for the department.

Sample Data (Same as above):

emp_id emp_name dept_id
1 Karthik 1
2 Ajay 1
3 Vijay 2
4 Vinay 2
5 Meena NULL

dept_id dept_name
1 HR
2 IT
3 Finance
Result:

emp_name dept_name
Karthik HR
Ajay HR
Vijay IT
Vinay IT
Meena NULL
- Note: Meena is included even though they don't belong to a department (NULL value in dept_name).

3. RIGHT OUTER JOIN (RIGHT JOIN)
A RIGHT JOIN returns all rows from the right table (departments), and the matching rows from the left table (employees). If no match is found, NULL is returned for the left table.
Example:
```
SELECT e.emp_name, d.dept_name
FROM employees e
RIGHT JOIN departments d ON e.dept_id = d.dept_id;
```
Explanation:
This query retrieves all departments and the corresponding employee names. If no employee belongs to a department, NULL is returned for the employee.
Sample Data (Same as above):
emp_id emp_name dept_id
1 Karthik 1
2 Ajay 1
3 Vijay 2
4 Vinay 2
5 Meena NULL

dept_id dept_name
1 HR

2 IT
3 Finance
Result:
emp_name dept_name
Karthik HR
Ajay HR
Vijay IT
Vinay IT
NULL Finance
• Note: The Finance department is listed even though no employees are in it (NULL for emp_name).

4. FULL OUTER JOIN
A FULL OUTER JOIN returns all rows from both tables, combining the results of LEFT JOIN and RIGHT JOIN. If no match is found, NULL is returned for the missing side.

Example:
SELECT e.emp_name, d.dept_name
FROM employees e
FULL OUTER JOIN departments d ON e.dept_id = d.dept_id;

Explanation:
This query retrieves all employees and all departments, even if there is no match. If there is no matching department for an employee, NULL is returned for dept_name, and vice versa.
Sample Data (Same as above):
emp_id emp_name dept_id
1 Karthik 1

emp_id emp_name dept_id
2 Ajay 1
3 Vijay 2
4 Vinay 2
5 Meena NULL

dept_id dept_name
1 HR
2 IT
3 Finance
Result:
emp_name dept_name
Karthik HR
Ajay HR
Vijay IT
Vinay IT
Meena NULL
NULL Finance
• Note: Both Meena (who is not assigned to any department) and the Finance department (which has no employees) are included.

30 QUESTIONS
Table Structure and Data (Example)
Here’s a sample table structure and data for these questions:

Table: employees
CREATE TABLE employees (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
manager_id INT,
dept_id INT
);
INSERT INTO employees (emp_id, emp_name, manager_id, dept_id) VALUES
(1, 'Karthik', NULL, 1),

```
(2, 'Ajay', 1, 1),
(3, 'Vijay', 1, 2),
(4, 'Vinay', 2, 2),
(5, 'Meena', 3, 3),
(6, 'Veer', NULL, 4),
(7, 'Keerthi', 4, 5),
(8, 'Priya', 4, 5);
```

Table: departments

```
CREATE TABLE departments (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50)

);

INSERT INTO departments (dept_id, dept_name) VALUES
(1, 'HR'),
(2, 'IT'),
(3, 'Finance'),
(4, 'Marketing'),
(5, 'Sales');
```

Table: projects

```
CREATE TABLE projects (
project_id INT PRIMARY KEY,
project_name VARCHAR(50),
emp_id INT
);

INSERT INTO projects (project_id, project_name, emp_id) VALUES
(1, 'Project A', 1),
(2, 'Project B', 2),
(3, 'Project C', 3),
(4, 'Project D', 4),
(5, 'Project E', 5);
```

1. Retrieve the names of employees and their corresponding managers from the "employees" table, ensuring that even employees without managers are included.
2. Display all employees and their corresponding departments from the "employees" and "departments" tables, showing employees even if they don't belong to any department.
3. List the names of employees who report to a manager, along with their manager's name, from the "employees" table.
4. Find the total salary paid to each employee and their respective department, including departments with no employees.
5. Display a list of employees who do not belong to any department, even if the department data is missing.
6. Fetch the names of employees and the projects they are assigned to. For employees who are not assigned any projects, show NULL for the project.
7. List all employees who have completed at least one project, showing their names and the project names.
8. Show the names of employees and their projects, ensuring that no project is omitted even if an employee is not assigned to it.
9. Find all employees and their corresponding salaries, and display NULL for salary if there is no salary record for the employee.
10. Retrieve the names of employees and their corresponding department names, including employees who are not in any department.
11. Find the names of all departments and employees, ensuring that departments with no employees are included.
12. List all employees with their contact information, including employees without contact records.
13. Show the names of employees and their department names, including employees not assigned to any department and departments without employees.
14. Find employees who have not completed any project, along with the project details where applicable.

15. Retrieve the names of employees and the names of their projects, including employees who are not working on any project.
16. List all projects and the employees assigned to them, even for projects that have no employees.

17. Show the names of all employees who have both a manager and at least one project, listing the manager's name as well.
18. List the names of employees and the corresponding department names, but exclude those employees who don't belong to a department.
19. Display employees who belong to multiple departments, showing the employee's name and the department names.
20. List the names of all departments and employees, ensuring that even if a department has no employees, it is included in the result.
21. Retrieve employees who have worked on at least one project and do not belong to a department, listing their name and project details.
22. Find the total number of employees who belong to a department, ensuring the departments with no employees are still included.
23. Show the employees and their managers, displaying only those employees who report to a manager, excluding employees without managers.
24. Display all employee names along with their corresponding managers' names, but include employees who do not have managers.
25. Find the names of departments and the number of employees in each department, including departments that have no employees.
26. List all employees and the departments they belong to, ensuring that departments with no employees are also listed.
27. Show a list of employees who do not have any corresponding salary records, along with their names.
28. Retrieve the names of employees and their project assignments, including employees who are not assigned to any projects.
29. List the names of all employees and their respective department and project assignments, including employees who are not assigned to a project or department.
30. Display the names of employees who belong to at least one department, with the department name listed, but include employees without a department as well